



Understanding Employee Stress in a Digital Workplace

Department of Business Administration

Business Analytics (BUSI 3341)

Section 201

Dr. Alabdulrahman, Rabaa

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Daman Arshad 202300481

Hussah Almogren 202100516

Executive Summary

Our main problem is employee stress and burnout which is linked to digital tool usage that is driving increased healthcare costs and turnover risk. Our current feedback is anecdotal; a data-backed strategy is needed to target the root causes of technology-related well-being decline. Work-related and social media screen time are the most critical, modifiable factors impacting employee mental well-being. Stress is higher in workers with higher work related screen hours.

We Recommend to Implement a “Smart Screen Use” Pilot Program targeting employees with high work-related screen time and launch “Digital Detox” Challenges & Location-Sensitive Flexibility to reduce social media anxiety and alleviate urban stress. This software-based intervention will prompt healthy usage habits, with a goal of reducing depression scores by 15% within one quarter. Incentivized team challenges and flexible arrangements aim to cut anxiety scores by 10% and lower urban stress by 5%.

These targeted initiatives address the most significant drivers of digital burnout. Successful implementation will lower healthcare utilization, reduce turnover risk, and enhance productivity, delivering a strong return on investment in employee well-being.

Report

Background

Our organization employs a diverse workforce increasingly reliant on digital tools for both work and personal life. With the rise of hybrid and remote work, the lines between professional and personal technology use have blurred. Rising healthcare costs and employee feedback surveys indicate stress and burnout are primary concerns affecting performance. This project aims to move beyond anecdotal evidence to identify specific, modifiable technology-related behaviors that influence employee well-being.

Role and Motivation

As the internal analytics team, our goal is to transform employee well-being data into actionable business strategy. Addressing technology-induced stress is critical not only for ethical employer responsibility but also for financial performance: stressed employees show 50% higher healthcare utilization and are 2× more likely to seek other employment.

Specify the Question

For our research, we identified five questions we want to look for and came up with a hypothesis for each.

H₁: Individuals who have higher daily screen time will report significantly higher stress levels compared to those with lower daily screen time.

H₂: Stress levels differ significantly between genders.

H₃: Stress levels significantly differ based on location type (urban, suburban, rural).

H₄: Higher social media screen time is associated with higher anxiety levels.

H₅: Participants with longer work-related hours will report higher depression scores.

Obtaining Data

Our data was sourced through Kaggle and contains 5000 individual records, obtained through a survey.

Our dataset was already mostly clean and well structured, thus we spent less time on analytical readiness. One step we took was fixing the formatting of the numbered data, including decimal points such as for phone usage. Next we added dummy variables as categorical encoding for our textual data to ensure that the different categories, including gender and location type, would hold the same weight in our calculations, such as for regression analysis and correlation test.

Furthermore, to ensure that the dataset was good, we used a correlation test, as shown in Figure 1, to check if the attributes were related to our research, which is about stress of individuals. We tried a regression test, however, it was not compatible with our dataset, as shown in Figure 2, in addition to the fact that we had more columns than allowed for a regression test in our Excel software.

Analyzing data

We employed a two-stage analytical approach, combining descriptive, diagnostic, and predictive analytics to understand the drivers of employee stress and well-being. For analyzing the data, we connected our dataset to another software known as Tableau. This allowed us to explore our data visually, and we could conduct descriptive analysis. We created interactive dashboards to explore relationships and test our hypotheses. Key analyses included:

1. **Comparison of Means:** We calculated and compared average stress across key groups such as gender, location, and screen time.
2. **Trend Analysis:** We visualized the relationship between continuous variables like daily screen time and average stress level.

Predictive Analytics (Regression Modeling): We attempted a multiple linear regression to model stress level. While the full model faced challenges, the diagnostic visual analyses provided statistically clear and actionable patterns to answer our core questions.

Limitations & Data Quality

Our initial regression model showed non-significance, likely due to multicollinearity among highly correlated screen-time variables (phone, laptop, daily total). However, the bivariate relationships visualized in our dashboards are robust and provide a strong foundation for business recommendations.

Results

H₁: **SUPPORTED.** Daily screen time is a major stress driver. The relationship is strong and positive: stress levels rise consistently as daily screen time increases, with a notable jump after 6+ hours.

H₂: **PARTIALLY SUPPORTED.** Gender shows a complex relationship with stress. While average stress levels are similar, the distribution differs very slightly: a higher proportion of males fall into the "High" stress category (20.26% vs. 19.72% for females).

H₃: **SUPPORTED.** Location significantly impacts stress. Urban employees report the highest average stress levels, followed by suburban and then rural. This confirms environmental context matters.

H₄: **SUPPORTED.** Social media usage is strongly linked to anxiety. Employees with high social media use (4+ hrs/day) have an average anxiety score 78% higher than low users (14.44 vs. 8.10). This is one of the strongest relationships found.

H₅: **SUPPORTED.** Work-related hours are a primary factor for depression. Employees with high work-related screen time (4+ hrs/day) have an average depression score nearly 60% higher than those with low work screen time (11.29 vs. 7.10).

Recommendations

Our first recommendation is to implement a **"Smart Screen Use" Pilot Program** to directly address the strongest drivers of stress and depression. Reducing passive and prolonged work-related screen exposure is likely to yield the greatest well-being ROI.

Action: Target employees with >6 hours of daily screen time or >4 hours of work-related screen time with software that prompts micro-breaks, enforces "focus blocks," and provides end-of-day screen time reports.

Success Metric: Aim for a 15% reduction in depression scores among the pilot group within one quarter.

Our second recommendation is to launch "Digital Detox" Challenges & Location-Sensitive Flexibility which will tackle the specific, high-impact issue of social media anxiety and the disproportionate stress burden on urban staff

Action: Run bi-monthly team-based challenges to reduce social media use during work hours, with incentives for the greatest reduction in anxiety scores. Offer additional flexible work arrangements (e.g., "no-meeting Wednesdays," remote work days) preferentially to urban-based employees to mitigate environmental stress.

Success Metric: Achieve a 10% reduction in average anxiety scores for participating teams and a 5% improvement in stress scores for urban employees opting into flexibility.

Future Considerations

To build on this analysis, we recommend:

1. **Advanced Segmentation:** Use cluster analysis (K-means) on the full 5,000-record dataset to identify distinct "employee well-being personas" (e.g., "The Anxious Social Media User," "The Burnt-Out Knowledge Worker") for hyper-personalized interventions.
2. **Causal Analysis:** Conduct an A/B test by rolling out the "Smart Screen Use" software to a randomly selected group and comparing their well-being metrics to a control group after 3 months.
3. **New Data Integration:** If possible, integrate anonymized calendar data (meeting density) and productivity tool logs to understand what type of screen activity (e.g., back-to-back video calls vs. focused writing) is most draining.

Report Appendix

	Female	Male	screen_time_usage	hsp_usage	hst_usage	h_usage	houal_media	hc_related	htainment	hming_hou	duration	lintake_mg	ly_anxiety	depression	stress_level
Female	1														
Male	-0.92491	1													
daily_scre	-0.01039	0.003595	1												
phone_usa	-0.00395	0.000628	0.48494	1											
laptop_usa	-0.01391	0.0106	0.54787	0.014591	1										
tablet_usa	0.012461	-0.00826	0.296935	0.00408	0.014786	1									
tv_usage_h	-0.01412	0.009728	0.492173	-0.00497	-0.01008	0.012382	1								
social_med	0.003843	-0.01394	0.634696	0.273316	0.299433	0.162047	0.276257	1							
work_relat	0.002032	-0.01386	0.55996	0.257368	0.267717	0.154612	0.254153	0.908517	1						
entertainm	-0.00591	0.015807	-0.50382	-0.20967	-0.2246	-0.12924	-0.22262	-0.70914	-0.67458	1					
gaming_ho	-0.00674	0.008655	-0.49492	-0.22584	-0.22645	-0.13088	-0.22541	-0.80057	-0.7628	0.573098	1				
sleep_dura	0.005954	0.003226	-0.41024	-0.21172	-0.25128	-0.1205	-0.1723	-0.2496	-0.22262	0.186953	0.200064	1			
caffeine_in	0.012822	-0.00522	0.076005	0.024898	0.057196	0.0162	0.026442	0.052592	0.047729	-0.05504	-0.04527	-0.19779	1		
weekly_an	-0.01089	0.004684	0.562312	0.219405	0.258173	0.143281	0.236697	0.808989	0.69873	-0.61815	-0.63082	-0.2131	0.056808	1	
weekly_dej	-0.00426	-0.00359	0.537885	0.208467	0.250976	0.130186	0.230485	0.772397	0.666602	-0.59057	-0.60257	-0.21926	0.054814	0.691101	1
stress leve	0.00114	-0.01046	0.665673	0.274263	0.297954	0.163577	0.281782	0.954274	0.826039	-0.73029	-0.74687	-0.26275	0.058176	0.848077	0.812952

Figure 1 Correlation Test, highlighting the relationship of stress and other factors

	A	B	C	D	E	F	G	H	I
1	SUMMARY OUTPUT								
2									
3	Regression Statistics								
4	Multiple R	0.97353718							
5	R Square	0.94777464							
6	Adjusted R Square	0.94768045							
7	Standard Error	0.66677634							
8	Observations	5000							
9									
10	ANOVA								
11		df	SS	MS	F	Significance F			
12	Regression	9	40261	4473.444	10061.939	0			
13	Residual	4990	2218.508	0.444591					
14	Total	4999	42479.51						
15									
16		Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
17	Intercept	9.94768261	0.135235	73.55857	0	9.682562867	10.2128	9.682563	10.2128
18	age	-0.00744588	0.000603	-12.3535	1.4852E-34	-0.008627506	-0.00626	-0.00863	-0.00626
19	Female	-0.08278928	0.049702	-1.66571	0.09583465	-0.180227439	0.014649	-0.18023	0.014649
20	Male	-0.12091391	0.049637	-2.43597	0.01488678	-0.218224079	-0.0236	-0.21822	-0.0236
21	daily_screen_time_hours	0.08663821	0.006833	12.6799	2.7591E-36	0.073243078	0.100033	0.073243	0.100033
22	physical_activity_hours_per_week	-0.03989659	0.006745	-5.9153	3.5335E-09	-0.053119061	-0.02667	-0.05312	-0.02667
23	mental_health_score	-0.05196915	0.001694	-30.6843	1.452E-189	-0.055289485	-0.04865	-0.05529	-0.04865
24	weekly_anxiety_score	0.08702504	0.003273	26.59046	8.296E-146	0.08060893	0.093441	0.080609	0.093441
25	weekly_depression_score	0.07920022	0.00328	24.14614	6.367E-122	0.072769909	0.085631	0.07277	0.085631
26	mood_rating	-0.47427513	0.007962	-59.566	0	-0.489884506	-0.45867	-0.48988	-0.45867

Figure 2 showing 0 significance for our model

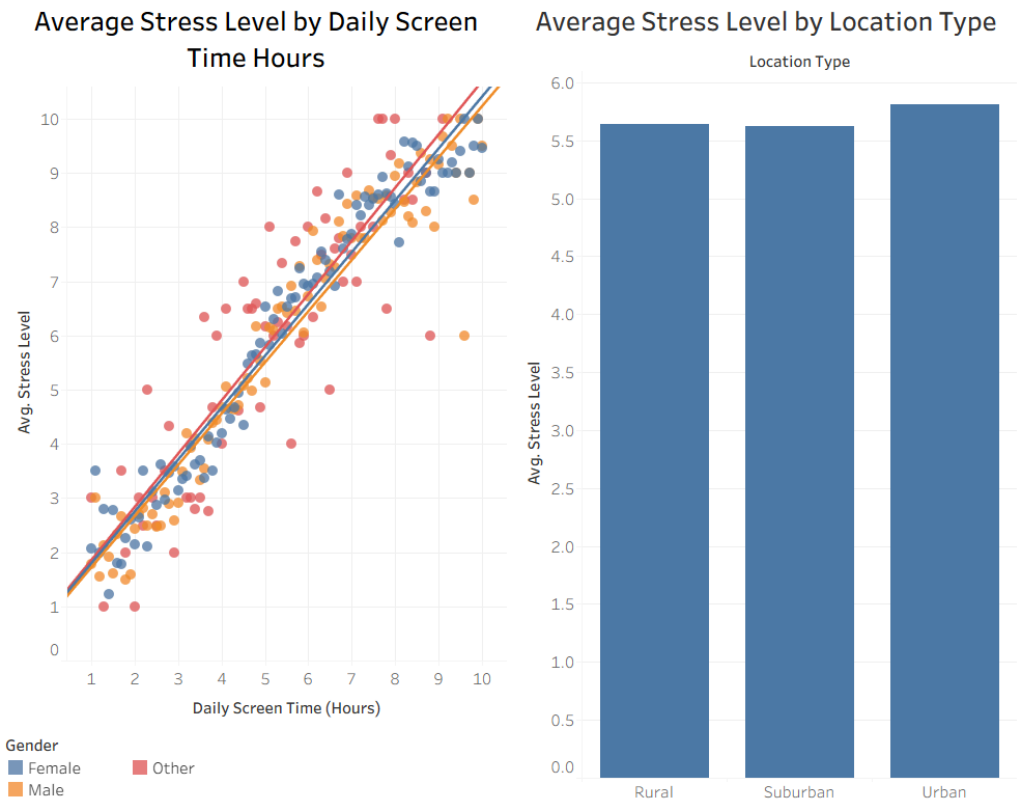


Figure 3 Dashboard part 1

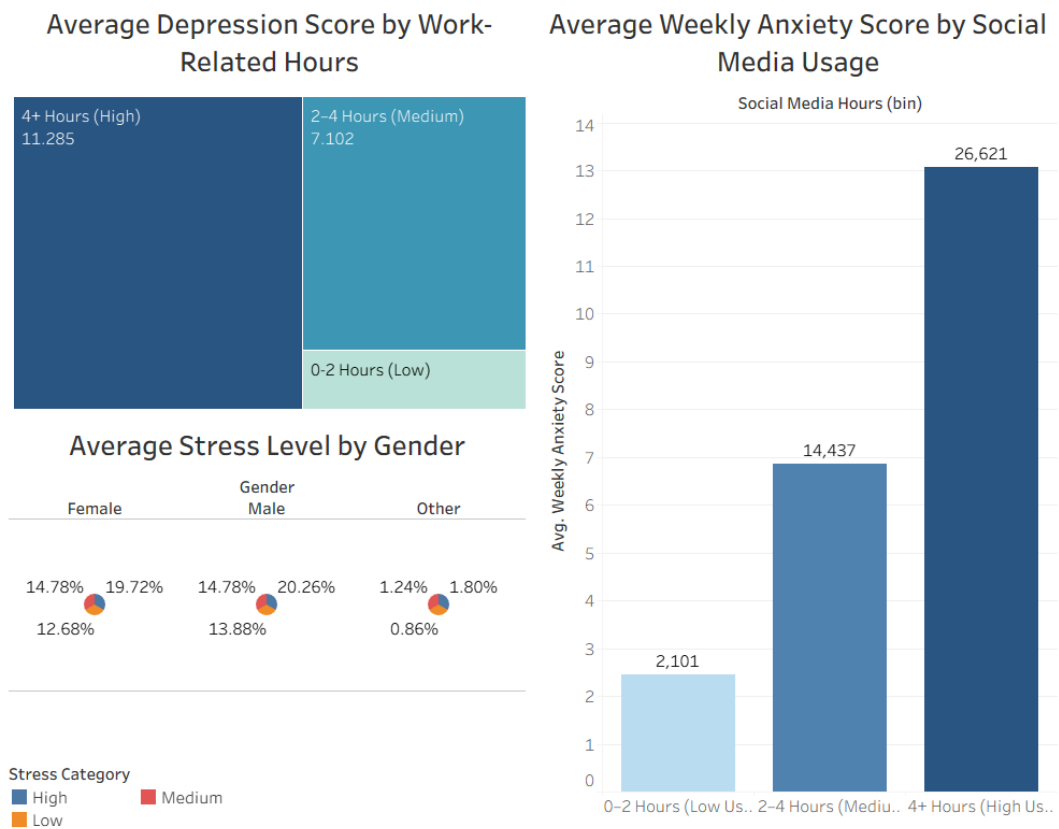


Figure 4 Dashboard part 2